RRO1 – EXACT SOURCES AND NORMALIZED REFERENCE RESPONSE

Abstract. This outline develops the exact external-source identities and the normalized reference-response estimate for the chamber QED3 regulator. It separates model-general zero-flux unfolding, source shifts and finite fourth-jet cancellation from the terminal operator, connected-logarithm and complement estimates. The endpoint concerns one fixed family of reference cells in the same harmonic-averaged zero-flux law, on a positive charge interval uniform in the ultraviolet cutoff.

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1. Introduction and target statement

RRO1 accompanies the 35-page manuscript RRM1 and is the exact-source and reference-response part of QEO1. It retains T05–T07, T12–T15 and Q4R. The quantitative theorem imports the selected trajectory from CVM1 and the terminal density and marked operator bounds from TPM1. The early exact identities are available independently of those response estimates.

Companion manuscript: RRM1, Theorem A. Overall plan: QEO1/Q4R. The boxes link to their full statement pages; every statement page links back to the overview.

Q4R. The normalized reference-response theorem

Model-specific. Overview Target.

Setting. Fix the chamber regulator and common trajectory specified in D, with $0 < \bar{m} \le 1$, and a real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$. For physical charge $e > 0$ and integer cutoff $N \ge 0$, let

$$h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \quad \eta = L^{-N}, \quad q = e\sqrt{\eta}, \quad K = N + \log_L h,$$
$$r_K = \min\{L^i : i \in \mathbb{Z}_{\ge 0},\ L^i \ge [-\log(e\sqrt{h})]^2\}.$$

and put

$$R_{\mathrm{top}} = hMr_K, \quad R^* = R_{\mathrm{top}}/L, \quad n_* = R^*/\eta, \quad \ell_c = R^*/(2L), \quad N_* = 8L^3.$$

For every $N \ge N_{\mathrm{ref}}(e)$, use the *same* real bare pair $(m_{0,N}, \epsilon^v_{0,N})$ selected on R_{top}, at every source and every twist. Write $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}$. This total mass may have either sign.

Let $\widehat{a}_c$, $c \in \mathcal{C}_* = (\mathbb{Z}/2L\mathbb{Z})^3$, be the oriented reflected probe copies in T05, at any admitted placement. They satisfy $\langle d\widehat{a}_c, d\widehat{a}_{c'} \rangle = v_\eta \delta_{cc'}$. On the unit torus $C(n_*)$, let γ_{n_*} be the standard

Gaussian on exact real two-cochains for the counting inner product, and let $A(u)$ be the mean-zero transverse potential of u. Set

$$H_{\theta,\nu} = \frac{\theta_\nu}{qn_*}, \qquad a(t) = e^{-1}\sum_c t_c \widehat{a}_c.$$

$$\mathring{\Phi}(t) = \frac{\displaystyle\int \frac{d\theta}{(2\pi)^3}\int d\gamma_{n_*}(u)\,\det\bigl(D^{\mathrm{ch}}[q(A(u)+H_\theta+a(t))]+\boldsymbol{\mu}_N\bigr)}{\displaystyle\int \frac{d\theta}{(2\pi)^3}\int d\gamma_{n_*}(u)\,\det\bigl(D^{\mathrm{ch}}[q(A(u)+H_\theta)]+\boldsymbol{\mu}_N\bigr)}.$$

Every twist integral is over $[-\pi,\pi]^3$. Its normalized measure is always $(2\pi)^{-3}d\theta$. The denominator is strictly positive. The free ratio instead uses the *positive reference mass*:

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[q(t/e)\widehat{a}_0]+\bar{m}\eta)}{\det(D^{\mathrm{ch}}[0]+\bar{m}\eta)}, \qquad c_m = N_0^{(m)}(0), \quad m=2,4.$$

These are ordinary derivatives of the ratio, not of its logarithm. Define

$$\Theta_m^{(0)} = \left|\left[\prod_{c\in\mathcal{C}_*}(\partial_{t_c}^m - c_m)\right]\mathring{\Phi}(0)\right|^{1/N_*}, \qquad \mathfrak{e} = \tfrac{1}{2}(1/4-9\varepsilon)-12\varepsilon \geq 17/400.$$

Claim. There is a positive e_{RR}, depending on the fixed construction, $\bar{m}$, and β, such that for every $0<e\leq e_{\mathrm{RR}}$, every $N\geq N_{\mathrm{ref}}(e)$, and every admitted placement, $\mathring{\Phi}$ is entire and

$$\Theta_4^{(0)} \leq Ce^{\mathfrak{e}}, \qquad (1+|c_2|)\Theta_2^{(0)} \leq Ce^{\mathfrak{e}}, \qquad C = 3\cdot 10^5(1+\sqrt{h}\,C_\beta)^4 h^{1/8}.$$

Here C_β is the finite charge-independent probe norm in T05. Every centered placement is admitted on this interval. Let $\mathbb{E}^{(0)}$ be the normalized zero-flux law on this *same reference torus*, with this same tuned mass, equivalently the above zero-source twist-averaged Gaussian density. For

$$X_c = \langle d\widehat{a}_c, F^{\mathrm{u}}\rangle, \quad H_2(y) = y^2 - v_\eta, \quad H_4(y) = y^4 - 6v_\eta y^2 + 3v_\eta^2,$$

all the following expectations are finite, and

$$\mathbb{E}^{(0)}\prod_c (e^{-m}H_m(X_c)-c_m) = \left[\prod_c(\partial_{t_c}^m - c_m)\right]\mathring{\Phi}(0), \qquad m=2,4.$$

Proof work. Identify the law and its exact source derivatives using T06 and T07. Factor the terminal density in T12; separate its one-cell errors from the connected several-cell logarithm in T13. The positive small-field representation and complex complement estimate of T14 supply the hypotheses of the finite polynomial cancellation theorem T15. Finally C verifies all numerical conditions on one cutoff-uniform positive charge interval. The number of differentiated variables is the fixed $N_* = 8L^3$, and the jet keeps degree four in *each* variable.

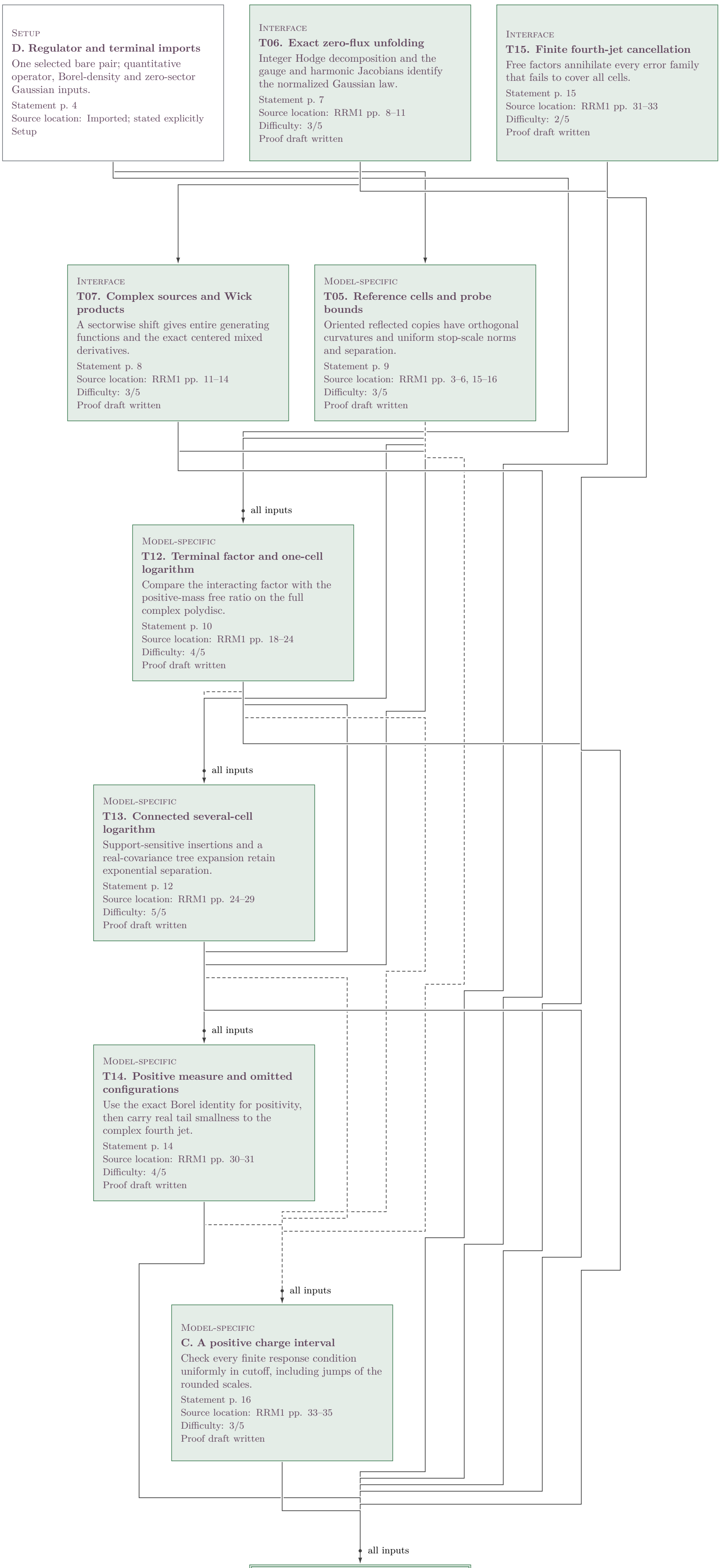
Inputs and endpoint. The quantitative terminal identities, locality bounds and the selected trajectory are the imports in D. The pressure conclusion of TPM1 and the positive continuum fourth coefficient of PRM1 are not inputs. Transfer to other volumes or flux sectors, continuum passage, and component cumulants are subsequent tasks in VTO1.

Source. RRM1, Theorem A, pp. 1–3; conditional estimate Theorem B, p. 32; Proposition 11.1, pp. 33–35.

RRO1 – Exact sources and normalized reference response

Proof draft written | Partial coverage | Not drafted

The double-bordered target is Q4R. D states the shared model and the quantitative imports; T06, T07 and T15 are general interfaces. The other result boxes concern the chamber regulator. The displayed length is the actual location in the 35-page RRM1 draft, not an additive page estimate.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

Conditional estimates feed C, which verifies their numerical hypotheses; C does not prove those estimates. T06 and T07 supply early identities usable by TPM1. TPM1's later terminal-density and operator outputs enter through D; its pressure conclusion is unused. Arrows are between these result ports, not whole manuscripts. Green means a proof draft is written, not that a new certification has been issued.

2. D. The regulator and the quantitative terminal imports

Setup. Overview Target.

Regulator. All finite cochain spaces have their counting inner products, extended *bilinearly* over $\mathbb{C}$. A reversed bond carries the negative cochain value. Let σ_ν be the Pauli matrices, $\gamma_\nu = \sigma_3 \otimes \sigma_\nu$, $\mathcal{O} = O(3) \cap \mathrm{GL}(3,\mathbb{Z})$, and

$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6), \quad 0 < \rho < 1,$$
$$(n_0, n_1, n_2, n_3) = (e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1).$$

The maps s_0, s_1, s_2 reverse coordinate three and interchange coordinates $(2,3)$ and $(1,2)$, respectively; r_{e_1} reverses coordinate one. Put $M_i(g) = w_i(\rho + \gamma \cdot g n_i)/2$. On four-spinors with labels $g \in \mathcal{O}$, antiperiodic in every coordinate,

$$\begin{aligned}(D^{\mathrm{ch}}[B]\psi)(x,g) = \sum_{i=0}^{2} M_i(g)(\psi(x,g) - \psi(x, gs_i)) \\ + M_3(g)(\psi(x,g) - e^{iB(x,x+ge_1)}\psi(x+ge_1, gr_{e_1})).\end{aligned}$$

Its argument is the *absorbed phase*. An unabsorbed unit connection A therefore occurs as $D^{\mathrm{ch}}[qA]$. There are 192 components per site. The chamber pairing has weight $1/48$; this constant does not rescale a finite determinant. Import the established finite positivity result: for every real connection and real lattice mass the determinant is nonnegative, and its harmonic average against the exact-curvature Gaussian is strictly positive. For complex connections it is entire, gauge invariant, $2\pi/q$-periodic under integer shifts, and bounded on every strip uniformly in the real part. Its degree in each of the two oriented link phases is at most 32.

The common construction. Fix the construction constants of QEM1, Proposition 4.1: $L \geq 13$ an odd integer, $M = L^m$, integer $m \geq 2$, $0 < \alpha < 1$, $1 - \alpha < \varepsilon \leq 1/200$, $r_{\mathrm{loc}} = 2$, $p_0 = 9$, $p = 19$, and positive ceilings $\bar{e}_1 \leq \bar{e}_0 < 1$. These ceilings are distinct from running charges. With $h, \eta, K, R_{\mathrm{top}}, R^*$ as in Q4R, import the common-pair theorem of CVM1. For

$$e \leq e_G := \begin{cases} \bar{e}_1, & h = 1, \\ \min\{\bar{e}_1/\sqrt{h}, (\sqrt{h}\log h)^{-1}\}, & h > 1, \end{cases}$$

it supplies a finite $N_{\mathrm{ref}}(e)$, one real pair selected on R_{top}, and its actual trajectory on R^*. For $N \geq N_{\mathrm{ref}}(e)$, the stop is $k_* \in \{K-1, K-2\}$, and

$$a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \eta_j = \eta/a_j, \quad \ell_j = R^*/a_j, \quad a_* \in \{h/L, h/L^2\}, \quad \ell_* = Mr_*.$$

Writing $v_j = -\log e_j$, the stopped activity and mass shift satisfy

$$|m_*| \leq \tfrac{1}{8} e_*^{3/4-8\varepsilon}, \qquad \|E_*\|_*^x \leq 4e_*^{1/4-6\varepsilon}.$$

This is the selected pair, not an arbitrary sequence with the same size bound. The positive mass in the blocked free operator is $\mu_* = \bar{m}a_*$; it is distinct from m_* and $\boldsymbol{\mu}_N$.

Operator and activity input. Import TPM1's terminal operator bounds and its marked derivative estimates. On coarse weight-one spinors, write $D_*(A)$ for the blocked free operator and $H_*^{\#}(A)$ for its fine reconstruction, including the covariant Hölder-pair rows and barred partners. Their fine-index weight is $\eta_*^3/48$, and their pair-index weight is its square. The positive block parameters satisfy

$$b_- = 1 - L^{-1} \leq b_* \leq 1, \quad \mu_- = L^{-3} < \mu_* \leq L^{-1}, \quad \gamma^F M \geq 16, \quad \ell_* > 2M + 2.$$

For real connections the block average obeys $Q_A Q_A^* = I$. For a bond field B, let $\|B\|_1, \|B\|_2, \|B\|_3$ denote its supremum norm, the supremum norm of its forward differences, and their α-Hölder seminorm over pairs at distance at most one. Put $\theta_i = e_j^{-3/4+i\varepsilon}$ and $R_j = \{B : \|B\|_i < \theta_i,\ i = 1,2,3\}$.

The background domain $\widetilde{R}_j$ consists of $B_0 + B_1$, where $B_1 \in R_j$, B_0 is real, and on each partition M-cube enlarged concentrically to side $(2L+1)M$, B_0 is real-gauge equivalent to a real field with these three bounds. The representatives may vary between cubes. The external domain is

$$\mathfrak{X}_j = \{C : |C|_\infty < e_j^{-1+2\varepsilon},\ |\partial C|_\infty < e_j^{-1+3\varepsilon}\}.$$

The total domain $\widetilde{G}_j$ consists of a real field locally real-gauge equivalent, on each partition M-cube enlarged concentrically to side $11M$, to a field of supremum norm below $e_j^{-1+\varepsilon}$, plus a complex field with the same supremum bound. Retain the imported inclusion $\widetilde{R}_j + \mathfrak{X}_j \subset \widetilde{G}_j$ on the admitted charge range. These are exactly RNM1's connection domains and total-connection domain. At the stop use $\mathcal{W} = \{A_r + C : A_r \in \widetilde{R}_* \text{ real},\ C \in \mathfrak{X}_*\}$. On this domain the inverse and reconstruction are analytic. Their localized majorants are $C^F \exp(-\gamma^F d(y,y'))$, including all short and long pair rows; long-pair distance is the minimum of the two endpoint-block distances. Their full component row sums are at most C^F. The first and mixed derivatives retain the intermediate support blocks: if P_B contains the endpoints of $\operatorname{supp} B$, a first coarse derivative is bounded by

$$226(C^F)^2 e_* |B|_\infty \sum_{z \in P_B} e^{-\gamma^F d(y,z)} e^{-\gamma^F d(z,y')};$$

mixed derivatives have the two ordered three-edge chains through the two supports. Field-leg derivatives have the corresponding chains through $P_B^+ = \{y : d(y,P_B) \le M+1\}$. These support factors are part of the import; row bounds alone do not suffice for T13.

Set

$$\mathcal{S} = C^F(24/\gamma^F)^3, \quad h_{\mathrm{L}} = 6\mathcal{S}, \quad C_H = 203M(C^F)^2, \quad C_K = 226\mathcal{S}^2, \quad C_K' = 1328\mathcal{S}^2,$$
$$c_\tau = \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \quad g = (b_-^{-1} + \mu_-^{-1})^{1/2}, \quad h_G = 1+g, \quad g_1 = b_-^{-1} + 4/\mu_-,$$
$$\theta_{\mathrm{CT}} = \mu_-/(12\mathrm{e}), \quad \vartheta_0 = \min\{\theta_{\mathrm{CT}}/2, \gamma^F/8, 1/M\}, \quad \alpha_G = 2b_-^{-1} + 4\mathrm{e}\mu_-^{-1}(24/\theta_{\mathrm{CT}})^3.$$

For every real A_r, $G = D_*(A_r)^{-1}$ has $\|G\|_2 \le g^2$ and component row and column sums weighted by $e^{\vartheta_0 d}$ at most α_G. In the complex extension $e_*|A - A_r|_\infty \le \mu_*/24$, whenever the fine inverse exists, $\|D_*(A)^{-1}\|_2 \le g_1$. The first derivative on the enlarged strip has constant C_K'.

For an even activity, its coefficient norm sums independent fine and pair generators with weights $h_1^n h_2^m/(n!m!)$, $h_1 = e_*^{-1/4}$, $h_2 = e_*^{-1/4+\varepsilon}$, and multiplies the polymer bound by $e^{\kappa d_M(X)}$. Here $M d_M(X)$ is the infimum of lengths of connected finite segment nets meeting every closed M-cube of X. Fix $\kappa \ge 7\kappa_0 + 7$, $\kappa_0 = 54\log 72$, and $K_0 = 72^{54}/36$, so the anchored polymer sum is at most K_0. Put $c_{\mathrm{rb}} = 54K_0$ and retain the fixed construction inequalities

$$L(\kappa - \kappa_0 - 1) \ge \kappa, \qquad c_{\mathrm{rb}} L^3 \exp(-[L(\kappa - \kappa_0 - 1) - \kappa]) \le 1/2.$$

The determinant activity is even, gauge invariant, local and analytic, with

$$|E_j^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}.$$

The common constant C_0 dominates the photon rows, fermion rows and Dirac form constants. In particular, with $c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}$, one requires

$$C_0 \ge \max\{1, C_P^{\mathrm{ph}} 3 e^{3\gamma_P} c(\gamma_P), C^F, 4 + 16 C^F c(\gamma^F)\}.$$

Measure input, independent of the response bound. TPM1 gives an exact Gaussian change of coordinates $F \mapsto (A_*; \zeta_j)_{j<k_*}$. The scalar ζ_j rows are orthonormal and number at most $2\ell_*^3 L^{3(k_*-j)}$; the $3\ell_*^3$ coarse plaquette rows have norm at most one. The coarse Gaussian m_* is nondegenerate on mean-zero coclosed real fields. Let

$$\mathfrak{G} = \{|\zeta_j| \le v_j^{p_0},\ |dA_*|_\infty \le v_*^p\}, \quad \chi_*(A) = \mathbf{1}_{\{|dA|_\infty \le v_*^p\}}, \quad c_{\theta,\nu} = \theta_\nu/(e_*\ell_*).$$

For every bounded Borel Ψ, every twist, and every admitted external source, its exact density identity is

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Psi(A_*)) = \int \Psi(A)\chi_*(A)Y_*^x(A + c_\theta; a^{[k_*]})\, d\mathsf{m}_*(A).$$

The density is jointly continuous on this support. Here $d\nu^{\theta,a}(F)$ is the exact-curvature Gaussian weighted by $e^{-\epsilon^v_{0,N} n_*^3} \det(D^{\mathrm{ch}}[q(A(F)+H_\theta+a)]+\boldsymbol{\mu}_N)/\det(D^{\mathrm{ch}}[0]+\bar{m}\eta)$. Writing $\bar{\nu}$ for its source-free twist average and $\bar{\mu} = \bar{\nu}/\bar{\nu}(1)$, the independent zero-sector estimate is

$$\bar{\mu}(e^{\langle f,F\rangle}) \le e^{\|f\|^2/2} \quad \text{for every real two-cochain } f.$$

It is a zero-sector trace estimate, not conditioning of an all-flux bound. Neither the pressure conclusion nor the response theorem is used in these inputs.

Source. RRM1, Section 1.1, p. 3; Section 5, pp. 17–18. Imports: CVM1, Theorem D and Proposition 6.1; TPM1, Proposition 2.1, Proposition 3.1, Theorem B, Section 6 and Appendix A; FBM1's determinant activity.

3. T06. Zero flux as a harmonic-averaged transverse Gaussian

Interface. Overview Target.

Objects and hypotheses. On the periodic cubical three-torus with sides $n_\mu \geq 4$, let $C^p(n;\mathbb{R})$ and $C^p(n;\mathbb{Z})$ be real and integer cochains, with the counting pairing and coboundary d. Put $V = n_1 n_2 n_3$, $q > 0$, $r_q = 2\pi/q$, $\Delta_0 = d^*d$, and

$$V_g = dC^0, \quad V_h = \{\text{constant one-cochains}\}, \quad V_t = (V_g \oplus V_h)^\perp, \quad F_n = dC^1.$$

For a closed integer two-cochain z, let $w(z) \in \mathbb{Z}^3$ be its sums over the coordinate two-tori. Let W be a bounded continuous complex function of real one-cochains, invariant under real exact shifts and r_q times integer shifts. For Borel Ψ with $|\Psi(u)| \leq C\exp(C\|u\|)$, define

$$Z_n^0[\Psi] = \sum_{\substack{z\in C^2(n;\mathbb{Z})\\ dz=0,\ w(z)=0}} \int_{[0,r_q)^{3V}} \Psi(dA + r_q z) e^{-\|dA+r_q z\|^2/2} W(A)\, dA.$$

Claim. Both sides of the following identity converge absolutely:

$$Z_n^0[\Psi] = \mathcal{N}_n \int \frac{d\theta}{(2\pi)^3} \int_{V_t} \Psi(dA) e^{-\|dA\|^2/2} W(A + H_\theta)\, d\sigma(A), \quad \mathcal{N}_n = r_q^{V+2} (\det{}'\Delta_0)^{1/2},$$

where $H_{\theta,\mu} = \theta_\mu/(qn_\mu)$, $\theta \in [-\pi,\pi]^3$, and σ is Euclidean Lebesgue measure for the counting pairing. In particular

$$\mathcal{G}_n := \int_{V_t} e^{-\|dA\|^2/2}\, d\sigma(A) = \frac{(2\pi)^{V-1}}{\det{}'\Delta_0}, \qquad Z_n^0[1] = \mathcal{N}_n \mathcal{G}_n z_{r,n},$$

$$z_{r,n} = \int \frac{d\theta}{(2\pi)^3} \int_{F_n} W(A(u) + H_\theta)\, d\gamma_n(u).$$

Here γ_n is the standard Gaussian on F_n, and $A(u) \in V_t$ is the unique potential. If W is nonnegative on real connections and $0 < Z_n^0[1] < \infty$, normalization gives the same probability law in both representations. The signed identity requires neither positivity nor a nonzero denominator.

Proof work. Fourier Hodge decomposition gives $\ker d|_{C^1} = V_g \oplus V_h$ and $\det(d^*d|_{V_t}) = (\det{}'\Delta_0)^2$. Separately prove the *integer* statement $\{z \in C^2(n;\mathbb{Z}) : dz = 0, w(z) = 0\} = dC^1(n;\mathbb{Z})$. An integer primitive is essential to unfold the compact bond cube. Closed integer one-cochains are exact integer cochains plus the three integer seam classes. Their coupling between exact and harmonic directions determines a fundamental domain. Its exact Gram determinant is $\det{}'\Delta_0/V$; its harmonic Jacobian is $V^{1/2}q^{-3}d\theta$. Combine them before converting to the normalized twist measure. Prove a Gaussian majorant before every unfolding or Fubini step.

Use. For the chamber determinant, real gauge and integer invariance follow from conjugation and its link phases. Its real strip bound supplies boundedness. Finite determinant positivity and strict positive harmonic average verify the separate probability hypotheses even at signed total mass. This application, not the general identity, uses D.

Source. RRM1, Sections 3.2–3.4, pp. 8–11; (30), (31), (38). Standard integer-cohomology input is cited there with its exact use.

4. T07. Complex source shifts and exact Wick products

Interface. Overview Target.

Objects and hypotheses. Use the finite cochain spaces, counting pairing, compact cube and flux vector of T06. Assume now that W extends to an entire function of complex one-cochains, invariant under complex exact shifts and real integer-period shifts, with a bound on each strip $|\operatorname{Im} A|_\infty \le s$ uniform in $\operatorname{Re} A$. For a fixed flux w, define $Z_n^w[\Psi]$ by the same sum and integral, with $w(z) = w$; define $Z_n^{x,w}(a)$ by replacing $W(A)$ by $W(A + a)$, leaving the photon Gaussian unchanged.
Claim. These external partition functions are entire in every finite family of complex sources. For every complex one-cochain a,

$$Z_n^w[e^{\langle da,F\rangle}] = e^{\langle da,da\rangle/2} Z_n^{x,w}(a).$$

The pairing in the exponent is bilinear, with no complex conjugation. The sectorwise identity may be summed over fluxes whenever the same strip-Gaussian majorant is used.

Suppose $W \ge 0$ on real connections and $Z_n^0[1] > 0$. For a finite index set $\mathcal{C}$, real sources a_c with $\langle da_c, da_{c'}\rangle = v\delta_{cc'}$, and a scaling constant $e > 0$, put

$$\Phi(t) = Z_n^{x,0}(e^{-1}\textstyle\sum_c t_c a_c)/Z_n^0[1], \quad X_c = \langle da_c, F\rangle.$$

For arbitrary real constants c_2, c_4, the normalized zero-flux expectation satisfies, for $m = 2, 4$,

$$\mathbb{E}^0 \prod_{c\in\mathcal{C}} (e^{-m} H_m(X_c) - c_m) = \left[\prod_c (\partial_{t_c}^m - c_m)\right] \Phi(0),$$

where $H_2(x) = x^2 - v$, $H_4(x) = x^4 - 6vx^2 + 3v^2$. Every polynomial moment appearing here is finite. Equivalently,

$$\mathbb{E}^0 \exp\left(\sum_c (z_c X_c - v z_c^2/2)\right) = \Phi(ez).$$

Proof work. For real a, complete the photon square and shift the compact fundamental cube. Integer representatives restore it, without changing the flux class, since the required change is an integer coboundary. The strip bound and Gaussian majorants justify the translated sums. Continue analytically in the finite source variables. Differentiate the normal-ordered generating function and then center by c_m. Orthogonality is what makes the normal-ordering exponential a product over cells. It does not assert independence of their interacting law.
Application. Q4R uses the chamber weight of D, the exact zero-flux identification T06, and the reflected family of T05. No RG estimate is needed for this identity. The same source shift is also used in the complement estimate of T14.

Source. RRM1, Proposition 3.1, p. 12; Proposition 3.2, pp. 13–14; (53).

5. T05. Probe cells and bounds at the reference stop

Model-specific. Overview Target.

Objects. Let β be a real compactly supported smooth two-form and $f = dd^*\beta$. Let D_β be the side of a least coordinate support cube. Enlarge it by two in each direction to a cube Q_β with center z_β and side $D_\beta^{\mathrm{buf}} = D_\beta + 4$. For $0 < \eta \le 1$, let

$$J_\eta u(y) = \eta^{-3} \int_{[-\eta/2,\eta/2]^3} u(y+z)\,dz, \quad \delta_{\eta,\mu} u(y) = \frac{u(y+\eta e_\mu/2) - u(y-\eta e_\mu/2)}{\eta},$$

and define the covering unit bond source

$$\widehat{a}^o_\nu(x) = -\eta^{1/2} \sum_{\mu\neq\nu} (\delta_{\eta,\mu} J_\eta \beta_{\mu\nu})(\eta x + \eta e_\nu/2).$$

Use the reference scales of Q4R. Set $w = n_*/(2L)$ and $Q_c^\circ = w(c + (0,1)^3)$. A placement $v \in \eta\mathbb{Z}^3$ is admitted if $v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3$. A placement is centered if $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2$. Translate β by v first. The map g_c acts in coordinate μ by $x \mapsto x + 2iw$ if $c_\mu = 2i$, and by $x \mapsto (2i+2)w - x$ if $c_\mu = 2i+1$. Transport oriented bonds, including their reversal signs, to obtain $(g_c)_* \widehat{a}^o$; then periodize on $C(n_*)$ to obtain $\widehat{a}_c$.

Claim. Under the placement and fit inequalities in C, the curvature supports lie strictly inside distinct cells, and

$$\langle d\widehat{a}_c, d\widehat{a}_{c'} \rangle = v_\eta \delta_{cc'}, \qquad v_\eta = \|d\widehat{a}^o\|_2^2.$$

The variance is independent of placement. If $a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu}$ and $f^\eta_{\mu\nu} = \delta_{\eta,\mu} a'_{\eta,\nu} - \delta_{\eta,\nu} a'_{\eta,\mu}$, then

$$d\widehat{a}^o(\sigma) = \eta^{3/2} J_\eta f^\eta(\text{physical center of } \sigma), \qquad \langle d\widehat{a}_c, F^{\mathrm{u}} \rangle = F^\sharp(\text{the reflected translate of } f^\eta).$$

Here $F^\sharp$ is the componentwise step field of value $\eta^{-3/2} F^{\mathrm{u}}$ on physical cubes. The exact source identity thus reads a smooth-test observable without a continuum limit.

Define connection rescaling by $a^{[j]}(b) = L^{j/2} a(L^j b)$, with $a_j = L^{j-N}$, $e_j = e\sqrt{a_j}$. At $a_* \in \{h/L, h/L^2\}$, put $\widehat{c}_c = e^{-1}\widehat{a}_c^{[k_*]}$. The external norm consists of the supremum of a connection, its first difference and its short Hölder first difference, at the two possible stops. Taking also all meshes and the coordinate reflections, it has a finite uniform bound C_β for the unscaled marks, and a finite energy bound $\mathbf{D}$, with

$$C_\beta \le 2\sqrt{h}(1 + h + 3h^2)|\beta|_{C^3}, \qquad v_\eta \le \mathbf{D} \le 48|\beta|^2_{C^2}(D_\beta^{\mathrm{buf}} + 3)^3.$$

Consequently $|a(t)|_{\mathrm{ext}} \le C_\beta |t|_\infty / e$. The endpoint supports of distinct stop-scale marks have distance at least $\ell_*/(8L)$. Let P_c be their endpoint unit blocks and $P_c^+ = \{y : d(y, P_c) \le M + 1\}$. Then

$$d(P_c, P_{c'}) \ge d_{\mathrm{sep}} := \frac{M r_*}{8L} - 1, \qquad |P_c^+| \le p_\beta := (D_\beta^{\mathrm{buf}}/a_* + 2M + 4)^3.$$

The loss one is needed when passing from endpoints to unit blocks.

Proof work. Use the centered-difference midpoint identity before periodizing. Track the support enlargement by $\eta/2$ per difference and the extra plaquette corners. Unfold one factor of the torus pairing; disjoint open cells leave only the equal-copy summands. For the uniform norms, write each centered difference as an average of an ordinary derivative and apply the same argument after rescaling. This proves finiteness of C_β and $\mathbf{D}$, rather than assuming unnamed placement estimates.

Inputs. Only the reference geometry and the two stop-scale identities in D; no response, positive continuum coefficient or pressure bound. The H^s approximation and variance convergence are recorded in RRM1, Section 2, but are not needed for this finite-reference endpoint.

Source. RRM1, Section 2, pp. 3–6, and Section 4, pp. 15–16, especially (58) and (59).

6. T12. The terminal factor and the free one-cell logarithm

Model-specific. Overview Target.

Objects. Use the terminal operator, common trajectory and measure inputs of D, and the separated marks of T05. Put

$$C_b = 1 + \sqrt{h}C_\beta, \quad t_x = e_*^{3\varepsilon}/C_b, \quad t_x^+ = e_*^{3\varepsilon}/(\sqrt{h}C_\beta), \quad r = t_x/4, \quad \mathscr{C}(t) = \sum_c t_c \widehat{c}_c,$$

where $t_x^+ = \infty$ when $C_\beta = 0$. For real stop fields with $|dA|_\infty \le p_* := v_*^p$, use the imported photon-domain estimate: the real gauge and its photon extension lie in the declared background domain, and the latter has component and derivative norms less than $e_*^{-\varepsilon}/2$. Choose a real axial gauge A^g with $|A^g|_\infty \le 2\ell_*^2 p_*$, and set $A_0 = \mathcal{H}_*^{\mathrm{ph}} A^g$, using the photon extension from the exact terminal density. It preserves real constants, so its twist is still $c_{\theta,\nu} = \theta_\nu/(e_*\ell_*)$.

Let $\mathcal{P}_j$ be the accumulated external-only determinant activity, defined by $\mathcal{P}_0 = 0$ and the actual reblocking recursion

$$\mathcal{P}_{j+1}(Y, C) = \sum_{\overline{X}^L = LY} (\mathcal{P}_j(X, C_L) + E_j^d(X, C_L)).$$

It obeys $|\mathcal{P}_j(X, C)| \le 2C_d M^3 e^{-\kappa d_M(X)}$ for $d_M(X) \ge 1$. Define the finite terminal integral

$$Z_{\mathrm{in}}(A, C) = \int D\Psi \, \exp\left[-\bar{\Psi} D_*(A + C)\Psi - m_*\langle \overline{H_*\Psi}, H_*\Psi\rangle + \sum_X E_*(X; A, C, H_*^{\#}\Psi)\right].$$

There are $4\ell_*^3$ weight-one coarse pairs. The reconstruction in the mass term and activity is at the indicated total connection. Put $Z_{\mathrm{tm}}(t) = Z_{\mathrm{in}}(A_0 + c_\theta, \mathscr{C}(t))$ and

$$f_{\mathrm{int}}(t) = \frac{Z_{\mathrm{tm}}(t)}{\det D_*(c_\theta + \mathscr{C}(t))}, \qquad R_{\mathrm{tm}}(t; \theta, A) = e^{\sum_X \mathcal{P}_*(X, \mathscr{C}(t))} \frac{Z_{\mathrm{tm}}(t)}{Z_{\mathrm{tm}}(0)}.$$

Quantitative hypotheses. Require $r_* > 31L + 16$ and

$$4C_0\ell_*^3 g_1 \le e_*^{-1/4+\varepsilon}, \quad 4\ell_*^3(2C_K + K_0 + 4) \le e_*^{-2\varepsilon}, \ e_*^{1/4-9\varepsilon} \le 1/8, \quad 48e_*^{2\varepsilon} \le \mu_-.$$

Define

$$C_{\mathrm{fr}} = 2592 C_d K_0 C_\beta^2 [M^2 (D_\beta^{\mathrm{buf}})^3 L^4 h^{-2} + 8M^5 h],$$
$$B_{\mathrm{wr}} = 16 \exp(6) C_d M^3 K_0 p_\beta e^{-r_*}, \quad B_1 = 16 g_1 C_K' p_\beta e_*^{2\varepsilon},$$
$$B_0 = C_{\mathrm{fr}} e_*^{4\varepsilon} + B_{\mathrm{wr}} + B_1, \qquad \tau_s = c_\tau \ell_*.$$

Also require

$$2g_1 C_K' e_*^{2\varepsilon} \le 1/2, \quad B_0 \le 1/4, \quad \tau_s \ge 1, \quad 52 B_1 e^{-\tau_s} \le e_*^{1/4-9\varepsilon}.$$

Claim. The external connections are admitted on the full polydisc $|t|_\infty < t_x^+$, and

$$|f_{\mathrm{int}}(t) - 1| \le \tfrac{1}{2} e_*^{1/4-9\varepsilon} \le 1/16.$$

The exact terminal ratio is nonzero and has a unique normalized analytic logarithm $\mathcal{G}(t; \theta, A)$, with $\mathcal{G}(0) = 0$, jointly continuous in the real parameters. Define

$$g_D^\theta(t) = \operatorname{tr} \log\{I + D_*(c_\theta)^{-1}[D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)]\}, \quad \ell_0(t) = \sum_X \mathcal{P}_*(X, t\widehat{c}_c) + g_D^0(t).$$

Then $\exp \ell_0 = N_0$, for the positive-mass free ratio in Q4R. This function is cell-independent, even and real on the real axis, and $|\ell_0(t)| \le B_0$ on $|t| \le t_x$. The one-cell error satisfies

$$w_c(t; \theta, A) := \mathcal{G}(t\mathbf{1}_c; \theta, A) - \ell_0(t), \qquad |w_c| \le 3e_*^{1/4-9\varepsilon} \quad (|t| < t_x^+).$$

For the fourth Taylor polynomial $\phi_0(t) = \sum_{j=0}^4 a_j t^j$ of N_0, set $\|\phi_0\|_r = \sum_{j=0}^4 |a_j| r^j$. Then

$$\|\phi_0\|_r \le 2, \qquad |c_2| \le r^{-2}/16, \qquad |c_4| \le r^{-4}/16.$$

Proof work. Telescope the fine determinant through the actual block integrations, keeping the 192-component incoming and four-component outgoing normalizations; only source-independent factors cancel. Compare the finite interacting integral with the free determinant using the terminal mass and activity bounds. Construct the normalized logarithm by integrating its logarithmic derivative on radial segments. For the free scalar activity, subtract a locally constant probe by a real gauge change and analytically continue the gauge parameter. Evenness gives a quadratic probe gain before summing the scales. Wrapping polymers cost e^{-r_*}. A Fourier contour shift in the twist variables gives $e^{-\tau_s}$, which controls the twist dependence of the one-cell logarithm. Cauchy estimates give the free jet bounds at the smaller radius.

Inputs. The exact external-slot terminal density, the common trajectory and operator/domain estimates in D; the finite probe norms in T05. This does not import the free continuum matching theorem.

Source. RRM1, Section 6, pp. 18–22, especially Proposition 6.1, p. 21; Section 7 and Proposition 7.1, pp. 22–24.

7. T13. A connected logarithm retaining the cell separation

Model-specific. Overview Target.

Objects and norms. Use $D_*, H_*^{\#}, E_*, m_*$, their support-sensitive derivatives and constants from D; use $t_x, p_\beta, \mathcal{G}$ and d_{sep} from T05 and T12. At a real background use the *real* covariance $G = D_*(A_r)^{-1}$, with $\|G\|_2 \le g^2$ and weighted row and column sums at most α_G. For an even coarse Grassmann polynomial F with zero constant coefficient, antisymmetric kernels F_p, and nonnegative symmetric weights w, set

$$\rho_G(F;w) = \sum_{p\ge 2} \frac{h_G^p}{(p-1)!} \sup_{u_1} \sum_{u_2,\dots,u_p} |F_p(u_1,\dots,u_p)| w(u),$$

$$N_G(F;w) = \sum_{p\ge 2} \frac{h_G^p}{(p-1)!} \sum_{u_1,\dots,u_p} |F_p(u_1,\dots,u_p)| w(u).$$

The index includes site, spin and barred/unbarred choice, all with weight one. The two weights used are $w_\delta(u) = e^{\vartheta_0 \max_{i,j} d(u_i,u_j)}$ and $w_P(u) = e^{\vartheta_0 \max_i d(u_i,P)}$. Absorb the mass shift into the activity, and write the terminal integral as $\det D_*(A_r)\, e^{V_0(t)} \int d\mu_G\, e^{V_+(t)}$, where

$$V(t) = -\bar\Psi[D_*(A_r + \mathscr{C}(t)) - D_*(A_r)]\Psi + \sum_X E_*^a(X;t)[H_*^{\#}(A_r + \mathscr{C}(t))\Psi],$$

V_0 is its scalar part and $V_+ = V - V_0$.

Hypotheses and insertion bounds. Besides T12, require

$$5h_{\mathrm{L}}h_G \le e_*^{-1/4+\varepsilon}, \qquad (4+M^3/2)e_*^{\varepsilon} \le 1, \qquad d_{\mathrm{sep}} \ge 6M+8,$$

$$\alpha_G \bar r \le 1/2, \quad \bar r = 4h_G^2 C_K e_*^{2\varepsilon} + 650 K_0 e_*^{1/4-7\varepsilon}.$$

Put $q_x = 2/t_x$, $s_F = (24/\gamma^F)^3$, and $C_V = 10^3 C_H^2 s_F^4 K_0$. The first task is to prove, uniformly in the background and twist,

$$\rho_G(V_+; w_\delta) \le \bar r \quad (|t|_\infty \le t_x),$$

$$N_G(\partial_c V_+; w_{P_c}) \le q_x C_V h_G^2 p_\beta, \qquad \rho_G(\partial_c V_+; w_{P_c}) \le q_x C_V h_G^2,$$

$$N_G(\partial_c\partial_{c'} V_+; 1) + |\partial_c\partial_{c'} V_0| \le q_x^2 C_V h_G^2 p_\beta e^{-\vartheta_0(d_{\mathrm{sep}}-4M-6)} \quad (c \ne c',\ |t|_\infty \le t_x/2).$$

Claim. The connected Gaussian series converges absolutely and gives the normalized logarithm $\mathcal{G}$. For a nonempty variable set $S \subset \mathcal{C}_*$, put

$$\mathcal{G}_S(t_S) = \sum_{T\subset S} (-1)^{|S\setminus T|} \mathcal{G}(t^T),$$

where t^T retains only coordinates in T. Every $\mathcal{G}_S$ vanishes if any variable of S is zero. For $|S| \ge 2$ and $|t|_\infty \le t_x/2$,

$$|\mathcal{G}_S| \le \varpi_{\mathrm{sep}} \le \varpi_{\mathrm{sep}}^+,$$

$$\varpi_{\mathrm{sep}} = 2^{N_*} C_W p_\beta e^{-\vartheta_0(d_{\mathrm{sep}}-4M-6)},$$

$$\varpi_{\mathrm{sep}}^+ = 2^{N_*} C_W p_\beta e^{-\vartheta_0(Mr_*/(8L)-4M-7)},$$

$$C_W = 3C_V h_G^2 + 2\alpha_G C_V^2 h_G^4 + 4\exp(6) C_d M^3 K_0.$$

Proof work. Differentiate the coarse quadratic term and every reconstructed activity leg, taking Cauchy estimates in the marked variable before the weighted polymer sum. Retain both first-derivative support weights; an unmarked norm would lose the separation. In the forest expansion, the interpolation matrix is positive semidefinite, including degenerate interpolation parameters. Its

Gram representation and $\|G\|_2 \le g^2$ bound the residual Wick determinant. The tree degree sum gives, when the weights pay a distance D,

$$|\mathcal{T}_n(F_1,\dots,F_n)| \le e^{-\vartheta_0 D}(n-2)!\alpha_G^{n-1} N_G(F_1;w_1)\prod_{i=2}^{n}\rho_G(F_i;w_i) \quad (n\ge 2).$$

The factorial $(p-1)!$ in the norms follows from summing tree degrees; it is not interchangeable with a coefficient supremum norm. Place two marks on separate vertices, or use the same-vertex mixed bound, to retain the path between their supports. Handle the scalar determinant activity separately: a contribution meeting two cells must pay the corresponding polymer diameter. Integrate the mixed derivative in the two coordinates and apply the variable-set inclusion–exclusion formula.

Source. RRM1, Section 8, pp. 24–29; Proposition 8.1, p. 25; Proposition 8.2, p. 27; Proposition 8.3, p. 29.

8. T14. A positive small-field measure and a complex jet bound

Model-specific. Overview Target.

Objects and hypotheses. Use the exact Borel density and photon rows of D, the terminal logarithm of T12, and the separation bound of T13. Let $\mathfrak{G}^-$ decrease every threshold in $\mathfrak{G}$ by one. Set $p_{0*} = v_*^{p_0}$, $p_* = v_*^{p}$, and

$$\varpi_0 = 18\ell_*^3 e^{-p_{0*}/2} + 6\ell_*^3 e^{-(p_*-1)^2/(2C_0^2)},$$
$$t_2 = e\min\{N_*^{-1/2}, C_\beta^{-1}, (2\sqrt{N_*\mathbf{D}})^{-1}\}, \quad \varpi_L = \sqrt{\varpi_0}e^{\mathbf{D}/2}, \quad M_L = 2 + e^{\mathbf{D}/2},$$
$$\lambda_L = (\varpi_L^{1/3} M_L^{2/3})^{1/N_*} 5(2N_* r/t_2)^4, \qquad \bar\kappa_1 = 6e_*^{1/4-9\varepsilon}.$$

A reciprocal of zero is interpreted as $+\infty$. Assume $v_* \geq 8$, $p_0 v_*^{p_0-1} \geq 28$, and, for some $0 < s < 1$,

$$2\varpi_{\mathrm{sep}}^{+} \leq \bar\kappa_1^{N_*}, \quad N_*\bar\kappa_1^{1-s} \leq 1/2, \quad \varpi_0 \leq 1/4, \quad 8N_* t_2 \leq t_x.$$

Claim. There is a *positive* measure v_{sf} on $[-\pi,\pi]^3 \times \{|dA|_\infty \leq p_*\}$, with mass between $1-\varpi_0$ and 1, for which

$$\frac{\bar\nu^{a(t)}(\mathfrak{G})}{\bar\nu(1)} = \int e^{\mathcal{G}(t;\theta,A)}\, dv_{\mathrm{sf}}(\theta, A).$$

The complement $L(t) = \bar\nu^{a(t)}(\mathfrak{G}^c)/\bar\nu(1)$ is entire. For its jet retaining every exponent at most four,

$$\|L^{[4]}\|_r := \sum_{\alpha_c \leq 4} |L_\alpha| r^{|\alpha|} \leq \lambda_L^{N_*}.$$

Proof work. Use the zero-sector Gaussian estimate on each photon row and sum the actual level counts to obtain $\bar\mu((\mathfrak{G}^-)^c) \leq \varpi_0$. Define v_{sf} by the source-free terminal density. Its positivity follows from the Borel identity for every nonnegative coarse observable, continuity of the density, and nondegeneracy of the coarse Gaussian. A negative or nonreal value on the interior would contradict that identity on a small ball; reach the boundary by radial continuity. Positivity must not be inferred from a formal Grassmann expression.

For real $|t|_\infty \leq t_2$, apply T07. Orthogonal probe curvatures make the shifted row displacements at most one, so the shifted omitted set lies in $(\mathfrak{G}^-)^c$. Cauchy–Schwarz and the Gaussian bound imply $0 \leq L \leq \varpi_L$. The strip-source bound gives $|L| \leq M_L$ on the complex t_2-polydisc. The finite-dimensional two-constants estimate then gives $\sup_{|t|_\infty \leq t_2/(2N_*)} |L(t)| \leq \varpi_L^{1/3} M_L^{2/3}$. Its proof uses the harmonic measure of a half-disc and iterates the coordinates; the exponent is independent of N_*. Finally apply Cauchy estimates and sum all coefficients with $\alpha_c \leq 4$. The resulting radius loss is precisely the factor in λ_L, including the fourth power per variable.

Inputs. The Borel identity, calibrated Gaussian rows and zero-sector domination in D; the exact complex source identity T07; the analytic nonvanishing and logarithm bounds of T12. The displayed separation and family-sum conditions are numerical inputs verified later in C.

Source. RRM1, Section 9, pp. 30–31; Proposition 9.1.

9. T15. A finite fourth-jet cancellation theorem

Interface. Overview Target.

Objects and hypotheses. Let $\mathcal{C}$ be any finite nonempty set of size n, $r > 0$, $0 < \kappa < 1$, $0 < s < 1$, and $\lambda \geq 0$. In the ring

$$\mathcal{J} = \mathbb{C}[t_c : c \in \mathcal{C}]/(t_c^5 : c \in \mathcal{C}), \qquad \|P\|_r = \sum_{\alpha_c \leq 4} |P_\alpha| r^{|\alpha|},$$

suppose ϕ_0 is an even one-variable polynomial with constant term one, $c_m = \phi_0^{(m)}(0)$, $m = 2, 4$, and

$$\|\phi_0\|_r \leq 2, \qquad |c_2| \leq r^{-2}/16, \quad |c_4| \leq r^{-4}/16.$$

Let v be a positive measure of mass at most one. For each nonempty $S \subset \mathcal{C}$, let k_S be a measurable $\mathcal{J}$-valued polynomial whose monomials use exactly the variables of S, each with positive exponent. Suppose, uniformly on the measure space,

$$\|k_{\{c\}}\|_r \leq \kappa, \qquad \|k_S\|_r \leq \kappa^n \ (|S| \geq 2), \qquad \|L\|_r \leq \lambda^n,$$

and

$$P = \prod_c \phi_0(t_c) \int \prod_{\varnothing \neq S \subset \mathcal{C}} (1 + k_S)\, dv + L.$$

Assume $n\kappa^{1-s} \leq 1/2$ and $\lambda \leq \kappa^s$.

Claim. For $m = 2, 4$,

$$\left| \left[\prod_c (\partial_{t_c}^m - c_m) \right] P(0) \right|^{1/n} \leq m! r^{-m} (2 \exp(1) \kappa^s + \lambda) \leq 8 m! r^{-m} \kappa^s.$$

The theorem concerns a fourth jet in every variable; it is not a bound on the total-degree-four truncation.

Proof work. The coefficient norm is submultiplicative. The centered derivative in cell c annihilates its free factor $\phi_0(t_c)$. Consequently only error families covering $\mathcal{C}$ survive. Charge cost one to a singleton set and cost n to every larger set. Every covering family has cost at least n. Keep κ^{sn}, and sum the remaining family weights using

$$(1 + \kappa^{1-s})^n (1 + \kappa^{(1-s)n})^{2^n - 1 - n} \leq \exp(1).$$

The norm of the tensor product of centered derivative functionals is at most $(m! r^{-m})^n$. The free factors cost 2^n, and the positive measure costs at most one. Treat L separately and take the n-th root.

Application. For Q4R, take $n = N_*$, $\kappa = \bar{\kappa}_1$, $\lambda = \lambda_L$, and $\phi_0 = N_0^{[4]}$. The singleton errors are $(e^{w_c} - 1)^{[4]}$, and the other errors are $(e^{\mathcal{G}_S} - 1)^{[4]}$. Cauchy estimates from radii t_x and $t_x/2$ to $r = t_x/4$ give their stated norms and variable supports. Joint continuity and Cauchy coefficient formulas justify moving the finite jet through the small-field integral.

Source. RRM1, Section 10, pp. 31–33, especially the representation before Theorem B and its covering-family proof. The general statement is the finite algebra used there, with no extension of its hypotheses.

10. C. Admission on a positive charge interval

Model-specific. Overview Target.

Objects and conditions. Fix the construction and mass of D, and the probe through only $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. Fix $s = 1/2$. The finite admission inventory comprises all displayed inequalities in T12, T13 and T14, and

$$r_* > 31L + 16, \quad v_* \geq 8, \quad p_0 v_*^{p_0 - 1} \geq 28, \quad M r_*/(8L) - 1 \geq 6M + 8, \quad \lambda_L \leq \bar{\kappa}_1^s,$$
$$D_\beta^{\mathrm{buf}}/2 + 1/2 \leq 3\ell_c/8, \qquad D_\beta^{\mathrm{buf}} + 3 \leq \ell_c.$$

In particular it includes the full-polydisc factor/mass conditions, $B_0 \leq 1/4$, twist error, weighted tree smallness, separated-cell error $2\varpi_{\mathrm{sep}}^+ \leq \bar{\kappa}_1^{N_*}$, family sum, photon tail and complex-radius conditions. All constants in this finite inventory are fixed before e.

Claim. There is $e_\sharp(\beta) > 0$, independent of N, twist and admitted placement, for which this entire inventory holds whenever $0 < e \leq e_\sharp$ and $N \geq N_{\mathrm{ref}}(e)$. Every centered placement is then admitted. Hence $e_{\mathrm{RR}} = \min\{e_G, e_\sharp\} > 0$ is a valid common interval in Q4R.

Proof work. With $v_K = -\log(e\sqrt{h})$, the two reference stops obey

$$v_K \leq v_*, \quad e_* = \exp(-v_*), \quad v_*^2 \leq r_*, \quad M v_*^2 \leq \ell_* < LM v_*^2,$$
$$p_\beta \leq (D_\beta^{\mathrm{buf}} L^2/h + 2M + 4)^3, \qquad \ell_c \geq hM v_K^2/(2L^2).$$

Volume factors are polynomial in v_*, whereas local smallness has positive exponential rate. Separation and wrapping errors have quadratic exponential decay. The potentially largest loss is the fourth-derivative radius in the complement. Establish explicitly

$$\varpi_0 \leq 24(LM)^3 v_*^6 e^{-v_*^2/2}, \qquad \lambda_L \leq A_8 \exp\left[-\frac{v_*^2}{12N_*} + \left(\frac{1}{N_*} + 4\right) v_*\right]$$

for a finite charge-independent A_8. These rates imply $\lambda_L \leq \bar{\kappa}_1^{1/2}$ on a sufficiently large half-line. The placement inequalities follow from $\ell_c \to \infty$. Finally define the admitted interval by requiring the entire inventory for *every smaller charge*, and take half the supremum of its nonempty set of upper endpoints. This handles jumps of the rounded stopping scales without asserting their monotonicity.

Inputs. Only the explicit reference-stop identities from CVM1 and the constants and numerical inventories on the preceding cards. The earlier cards are conditional estimates; this is the separate verification that their conditions can hold simultaneously.

Source. RRM1, Section 11, pp. 33–35; Proposition 11.1 and the final proof of Theorem A.